

# Yingrui Hou

## Data Scientist — Health Monitoring & Anomaly Detection (noisy sensor systems)

Target Application: Safran Analytics | Data Scientist Health Monitoring  
F/H

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### Introduction personnelle

Data scientist with a PhD background in experimental physics and several years building robust Python/C++ inference and validation workflows on noisy measurement systems.

Strength: turning imperfect, biased data into reliable indicators through calibration/correction, uncertainty-aware validation, and reproducible pipelines.

Best fit: industrial teams working with sensor/time-dependent data where model credibility and failure-mode analysis matter.

### Selected Fit For This Role

- Developed reusable Python/C++ workflows for validation, calibration, and analysis of noisy measurement systems in shared technical environments.
- Built monitoring and validation workflows to identify reconstruction and calibration discrepancies, then trace anomalies back to detector or software causes.
- Identified concrete anomalies that led to corrective actions, including a missing-energy software issue later fixed in reconstruction.
- Developed ROOT/C++ calibration and correction workflows to reduce structured bias and compare performance before and after correction across operating conditions.
- Worked in large shared codebases with review discipline, result reproducibility, and reinjection of validation outputs into modelling decisions.
- Aligned simulation and data through reweighting and control-sample calibration before downstream decisions, avoiding the propagation of software or model bias.

*software validation, automation, Python, Linux, critical environment, and a direct link to software quality and internal simulation tools.*

#### Education

PhD in Particle Physics  
University of Chinese Academy of Sciences  
BSc in Applied Physics  
China University of Mining and Technology

#### Languages

Chinese: Native speaker  
English: C1  
French: B1 to B2

### Core Tools & Methods

**Tools:** ROOT / RooFit | Python | NumPy | pandas | C/C++ | statsmodels | matplotlib / seaborn | Geant4 | Workflow Automation (YAML / Shell / CMake) | Scientific Software | etc

**Methods:** event-level correction | Probabilistic Modelling | Time-Series Modelling | Statistical Inference | Time-Resolved Signal Analysis | particle-dependent timing models | covariance-aware joint fitting | statistical background subtraction | Maximum Likelihood Estimation | Large-Scale Data Analysis | etc

**Validation:** Calibration & Bias Correction | Model Validation | Root-Cause Analysis | Bias Correction | resolution calibration | Simulation-to-Data Alignment | Simulation Calibration | Simulation Correction | Monitoring | Data Validation | etc

**ML:** Domain Adaptation | Reweighting | Feature Validation | gradient-boosting reweighting | scikit-learn | CatBoost / Gradient Boosting | GBDT | CatBoost | Feature Selection | CatBoost training | etc

### Relevant Experience

*Postdoctoral Researcher | 2021 - 2025*

### Inference, Validation & Analysis Workflows

*CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration*

- Built reusable C++/Python/ROOT workflows for weak-signal extraction, time-dependent inference, and validation on noisy datasets where direct event-level truth was unavailable.
- Implemented likelihood-based models and joint fits to estimate latent parameters under sample mismatch, finite resolution, and biased observation conditions.
- Aligned simulated and observed datasets through gradient-boosting reweighting and control-sample calibration before downstream inference and selection decisions.

## Simulation, Calibration & Test-Data Interpretation

CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration

- Combined beam-test data, simulation, and response maps to model measurement-chain behaviour and quantify non-uniformity effects on system resolution, including a quantified constant-term contribution below 1%.
- Built time-resolved signal-analysis workflows that recovered component fractions from scintillation timing and improved reconstructed-energy resolution by about a factor of 2 in representative PSD studies.
- Built benchmark-based monitoring workflows to validate reconstruction and calibration quality, then traced anomalies to detector behaviour or software issues instead of stopping at descriptive analysis.

### Selected Project

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#### Weak-Signal Inference

*Mixture Models | Maximum Likelihood | Feature Selection | Statistical Validation*

Built a reusable C++/Python/ROOT workflow for weak-signal extraction in data where background dominates and event-level truth is unavailable.

**Methods:** likelihood fitting, control-sample validation, production yield workflows

**Impact:** turned rare-signal extraction into a reproducible workflow with stability checks and auditable validation logic.

**Industry relevance:** Rare Event Analytics | Anomaly Detection | Weakly Labeled Systems

#### Time-Series Inference

*Time-Series Modelling | Bias Correction | Calibration | Joint Fitting*

Built a bias-corrected time-series inference workflow for data affected by selection effects, detector resolution, and sample mismatch.

**Methods:** acceptance modelling, resolution calibration, custom fit components, joint fitting

**Impact:** converted a bias-sensitive latent-dynamics problem into a controlled inference workflow with dataset-specific corrections.

**Industry relevance:** Durability Modelling | State Estimation | Latent-Dynamics Inference

#### Test-Data Validation & Root-Cause Analysis

*Monitoring | Data Validation | Performance Diagnostics | Root-Cause Analysis*

Built a production-style validation workflow for early Run 3 photon reconstruction, focused on finding actionable detector and software issues.

**Methods:** benchmark-channel checks, MC-data comparison, detector-level diagnostics

**Impact:** translated validation discrepancies into concrete calibration, detector, and software follow-ups before they became accepted bias.

**Industry relevance:** Production Monitoring | Model Validation | Observability | Quality Assurance